(FP1) 2. Rational functions and graphs 专题作业

NO. 1

Let a be a positive constant.

a. The curve C_1 has equation $y = \dfrac{x-a}{x-2a}$. [2]

Sketch C_1.

The curve C_2 has equation $y = \left(\dfrac{x-a}{x-2a}\right)^2$. The curve C_3 has equation $y = \left|\dfrac{x-a}{x-2a}\right|$.

b. i. Find the coordinates of any stationary points of C_2. [3]

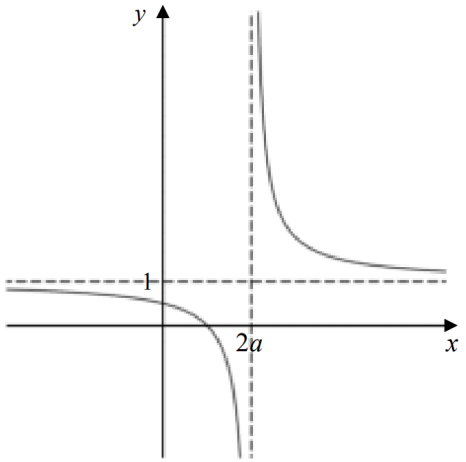
ii. Find also the coordinates of any points of intersection of C_2 and C_3 . [3]

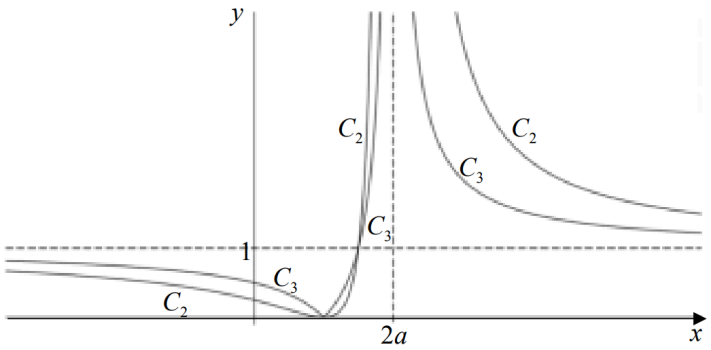
c. Sketch C_2 and C_3 on a single diagram, clearly identifying each curve. Hence find the set of values of x for which $\left(\dfrac{x-a}{x-2a}\right)^2 \leqslant \left|\dfrac{x-a}{x-2a}\right|$. [5]

A level FURTHER MATHEMATICS 2020/Winter/Paper1_2/Q6

Mark scheme

Question	Answer	Marks	Guidance

a		B1	Asymptotes labelled.
		B1	Correct position and shape. Not too truncated.
		2	
b.i	$\frac{dy}{dx} = -\frac{2a(x-a)}{(x-2a)^3} = 0$	**M1**	Sets $\frac{dy}{dx} = 0.$
	Solves for x.	**M1**	
	Since $x \neq 2a$, stationary point is$(a,0)$.	**A1**	
		3	
b.ii	$(x–a)2 = (x–a)(x–2a)$ or $(x–a)2 = –(x–a)(x–2a)$	**M1**	Finds a critical point.
	$x=a$ and $\frac{3}{2}a$ **or** $(a, 0)$ **or** $(\frac{3}{2}a, 1)$	**A1**	Both x values or one correct point.
	$(a, 0)$ and $(\frac{3}{2}a, 1)$	**A1**	
		3	

c		**B1**	Axes and asymptotes.
		B1 FT	C_3 FT from their sketch in part **(a)**. Clearly identified.
		B1	Correct shape of C_2. Clearly identified.
		B1	Relative positions and intersections correct.
	$x \leq \frac{3}{2}a$	**B1**	Accept algebraic method.
		5	

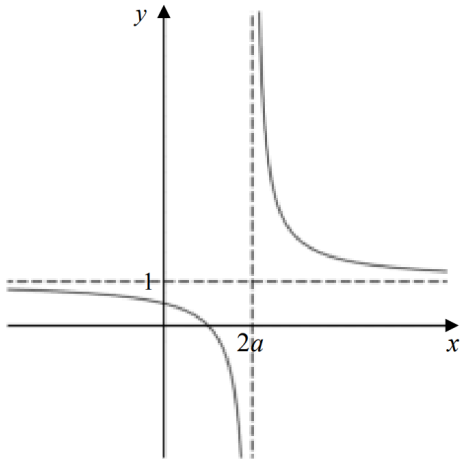
Explanation

a.

Convert C1 to a proper fraction:

$$y = \frac{x-2a+a}{x-2a} = 1 + \frac{a}{x-2a}$$

This is a reciprocal function followed by a translation. Asymptotes are x=2a, y=1.



B1	**Asymptotes labelled.**
B1	**Correct position and shape. Not too truncated.**

b.i

Stationary point: $\frac{dy}{dx} = 0$

$\frac{dy}{dx} = 2\left(\frac{x-a}{x-2a}\right) \cdot \frac{(x-2a)-(x-a)}{(x-2a)^2} = \frac{-2a(x-a)}{(x-2a)^3}$

$\frac{-2a(x-a)}{(x-2a)^3} = 0$ **M1**

$x = a$ **M1**

When x=a, y = 0

Stationary point: (a,0) **A1**

ii) solve simultaneous equations to find intersections:

$\left(\frac{x-a}{x-2a}\right)^2 = \left|\frac{x-a}{x-2a}\right|$

Square both sides:

$\frac{(x-a)^4}{(x-2a)^4} = \frac{(x-a)^2}{(x-2a)^2}$

Multiply by $(x-2a)^4$, since $x \neq 2a$

$(x-a)^4 = (x-a)^2(x-2a)^2$

$(x-a)^2((x-a)^2-(x-2a)^2)=0$ **M1**

$(x-a)^2(2x-3a)a=0$

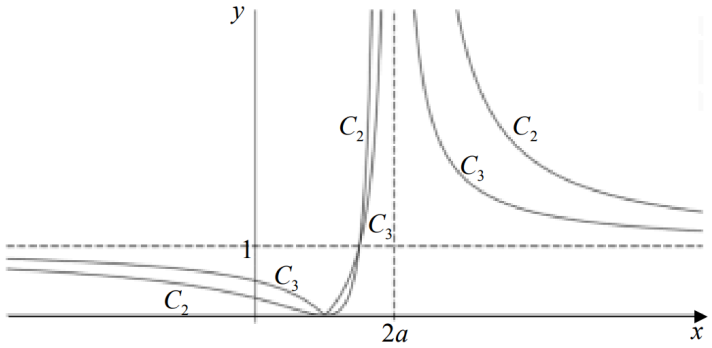
$x=a \text{ or } x=\frac{3}{2}a$ **A1**

When $x=\frac{3}{2}a$, y=1

Intersections: (a,0) $\left(\frac{3}{2}a,1\right)$ **A1**

c.

Modulus is to reflect in x -axis the part of the curve below x-axis while square the function takes one more step, decreasing the values below y=1 and increase the value above y=1.



B1	**Axes and asymptotes.**
B1 FT	**C_3 FT from their sketch in part (a). Clearly identified.**
B1	**Correct shape of C_2. Clearly identified.**
B1	**Relative positions and intersections correct.**

From graph, solution set is where C2 curve is below the C3curve:

Critical point is $\left(\frac{3}{2}a, 1\right)$

Solution set: $x \leq \frac{3}{2}a$ **B1**

NO. 2

The curve C has equation $y = \frac{x^2+x-1}{x-1}$.

a. Find the equations of the asymptotes of C. [3]

b. Show that there is no point on C for which $1 < y < 5$. [4]

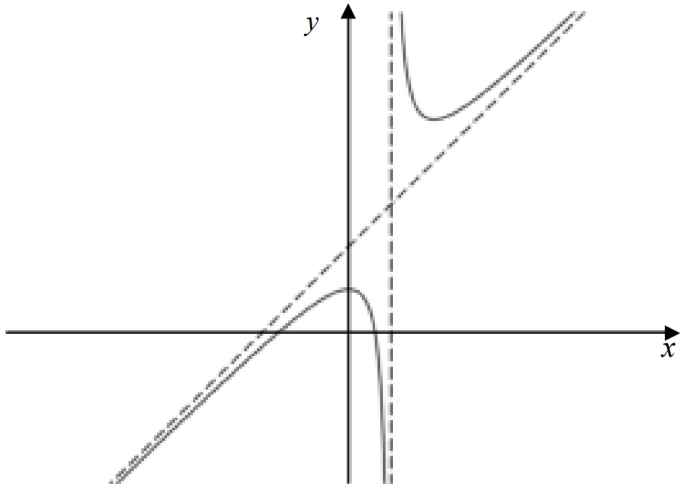
c. Find the coordinates of the intersections of C with the axes, and sketch C. [3]

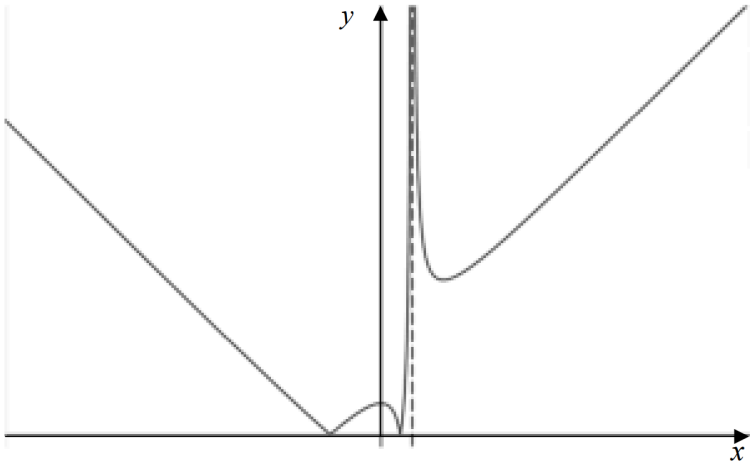
d. Sketch the curve with equation $y = \left|\frac{x^2+x-1}{x-1}\right|$. [2]

A level FURTHER MATHEMATICS 2020/Winter/Paper1_1/Q6

Mark scheme

Question	Answer	Marks	Guidance
a	$x = 1$	**B1**	States vertical asymptote.
	$x^2+x-1 = (x-1)(x+2)+1 \Rightarrow y=x+2$	**M1 A1**	Finds oblique asymptote.
		3	
b	$yx - y = x^2 + x - 1 \Rightarrow x^2+(1-y)x + y - 1 = 0$	**M1 A1**	Forms quadratic in x.

	$(1-y)^2 - 4(y-1)<0 \Rightarrow y^2 - 6y + 5 < 0$	**M1**	Uses that discriminant is negative if there are no values of x.
	$1 < y < 5$	**A1**	AG
		4	
c		**B1**	Axes and asymptotes.
		B1	Branches correct.
	$(0,1), \quad \left(-\frac{1}{2}+\frac{1}{2}\sqrt{5},0\right), \quad \left(-\frac{1}{2}-\frac{1}{2}\sqrt{5},0\right)$	**B1**	States coordinates of intersections with axes, can be labelled on graph. Accept (0.618,0) and (−1.62,0).
		3	

d		B1 FT	FT from sketch in **(c)** with both branches.
		B1	Correct shape at extremities.
		2	

Explanation

a.

vertical asymptote: x -1 = 0 -> x=1 **B1**

Convert to proper fraction:

$$y = \frac{x^2+x-1}{x-1} = \frac{x^2-x+2x-2+1}{x-1} = x + 2 + \frac{1}{x-1}$$

Oblique asymptote: y = x+2 **M1 A1**

b)

Convert the original to a quadratic respect to x:

$$y(x-1) = x^2 + x - 1$$

$$x^2 + (1-y)x + (y-1) = 0$$ **M1 A1**

When x has no solutions:

$$\Delta = (1-y)^2 - 4(y-1) < 0$$

$y^2 - 6y + 5 < 0$ **M1**

1<y<5 **A1**

c)

intersection with y-axis:

x=0 -> y=1

(0,1)

Intersections with x-axis:

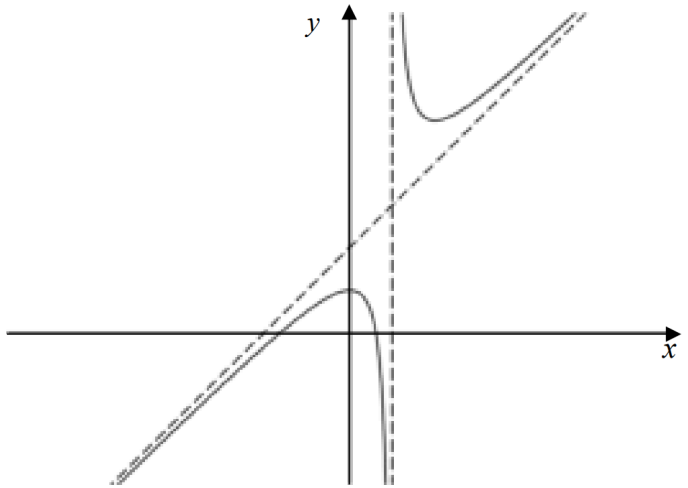
y=0 ->

$x^2 + x - 1 = 0$

$x = \frac{-1 \pm \sqrt{5}}{2}$

$\left(\frac{-1-\sqrt{5}}{2}, 0\right), \left(\frac{-1+\sqrt{5}}{2}, 0\right)$ **B1**

When x >1, y>3. So two branches are top-right and bottom-left.



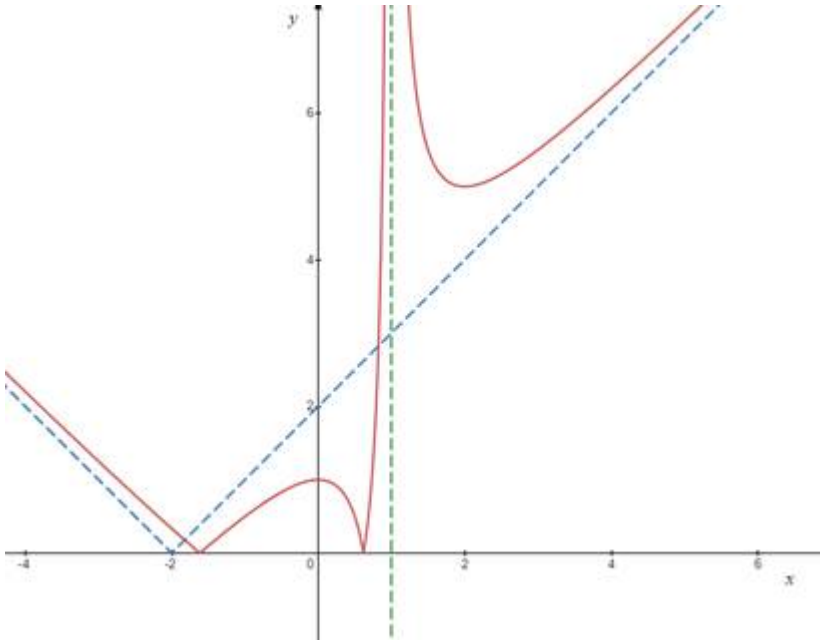
B1	**Axes and asymptotes.**
B1	**Branches correct.**

B1	**States coordinates of intersections with axes, can be labelled on graph. Accept (0.618,0) and (−1.62,0).**

d.

Reflect in x -axis the part of the curve below x-axis and get the graph.

Notice that asymptotes and intersections have the same reflection.



B1 FT	**FT from sketch in (c) with both branches.**

B1	**Correct shape at extremities.**

NO. 3

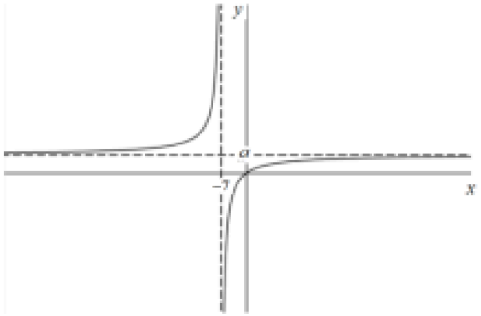
Let a be a positive constant.

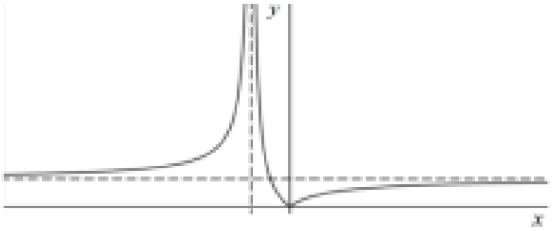
a. Sketch the curve with equation $y = \frac{ax}{x+7}$. [2]

b. Sketch the curve with equation $y = \left|\frac{ax}{x+7}\right|$ and find the set of values of x for which $\left|\frac{ax}{x+7}\right| > \frac{a}{2}$. [4]

A level FURTHER MATHEMATICS 2020/Summer/Paper1_1/Q1

Mark scheme

Question	Answer	Marks
a	 (B1 for axes and asymptotes correct, B1 for branches correct)	**B1** **B1**

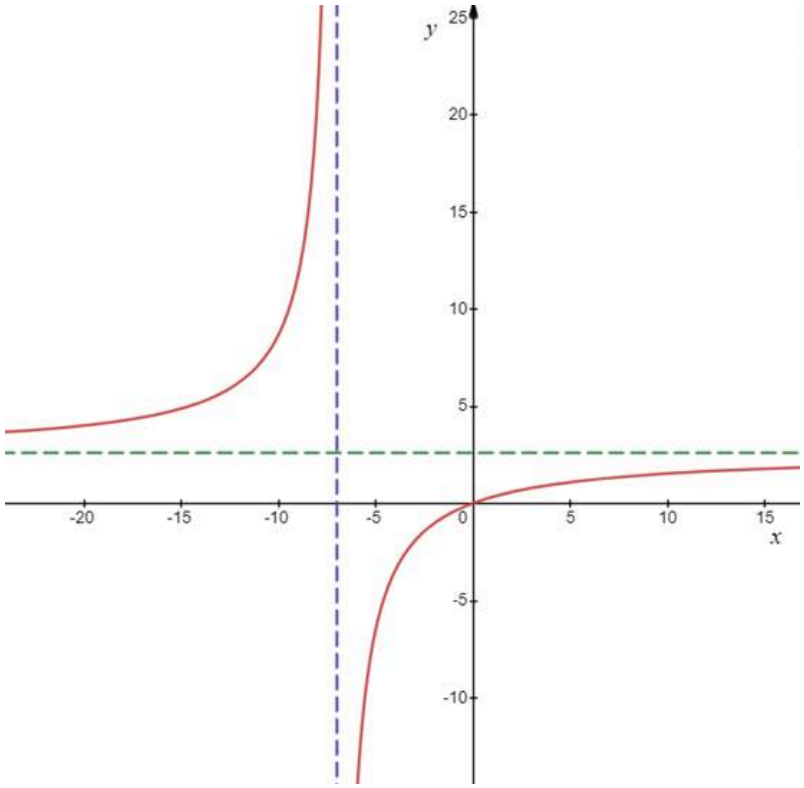
		2
b	 (B1 FT from sketch in part (a))	**B1 FT**
	$\frac{ax}{x+7}=\frac{a}{2}$ or $\frac{ax}{x+7}=-\frac{a}{2}$	**M1**
	$x=7$ and $x=-\frac{7}{3}$	**A1**
	$x<-7$, $-7<x<-\frac{7}{3}$, $x>7$	**A1**
		4

Explanation

a.

$$y=\frac{ax+7a-7a}{x+7}=a-\frac{7a}{x+7}$$

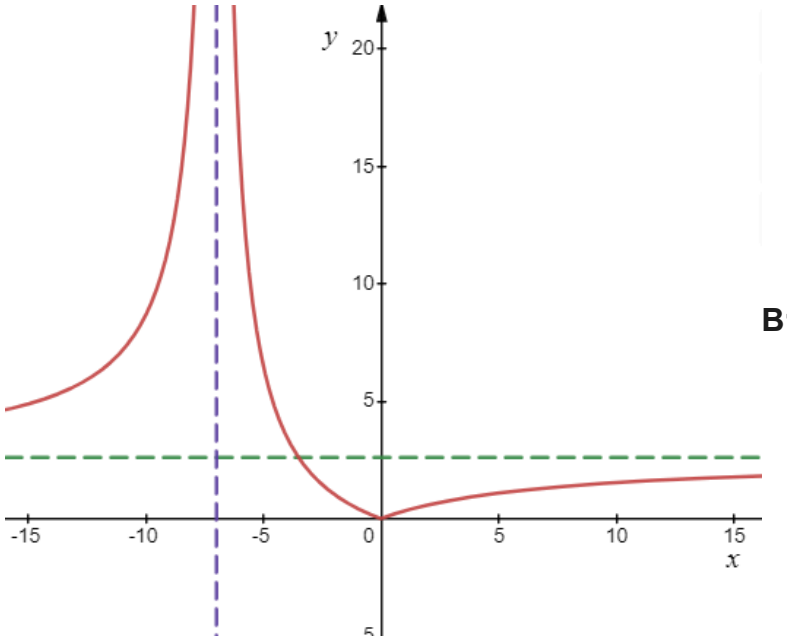
Asymptotes: x =-7, y=a



(B1 for axes and asymptotes correct, B1 for branches correct)

b)

Reflect in x -axis the part of the curve below x-axis and get the graph.



B1

Find critical points:

$\frac{ax}{x+7} = \frac{a}{2}$ *or* $\frac{ax}{x+7} = -\frac{a}{2}$ **M1**

$2x = x + 7,\ 2x = -x - 7$ (a>0)

$x = 7, x = -\frac{7}{3}$ **A1**

Since x ≠-7, solution set is $x < -7, -7 < x < -\frac{7}{3} or\ x > 7$ from graph. **A1**

NO. 4

The curve C has equation

$$y = \frac{x^2 + ax - 1}{x + 1},$$

where a is constant and $a > 1$.

i. Find the equations of the asymptotes of C. [3]

.....

ii. Show that C intersects the x-axis twice. [1]

.....

iii. Justifying your answer, find the number of stationary points on C. [2]

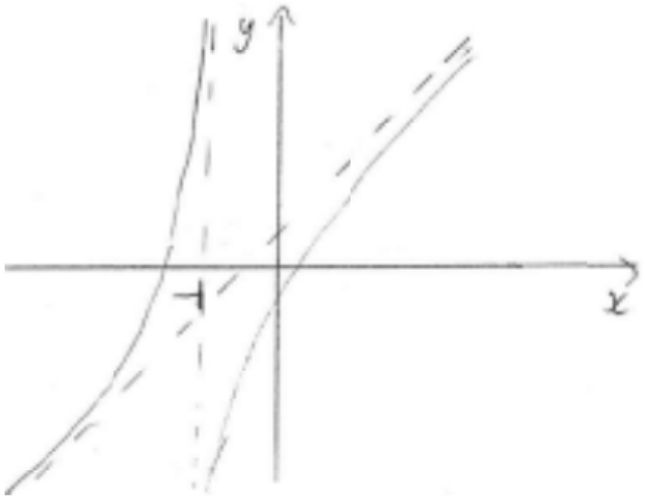
.....

iv. Sketch C, stating the coordinates of its point of intersection with the y-axis. [3]

A level FURTHER MATHEMATICS 2018/Winter/Paper1_1/Q6

Mark scheme

Question	Answer	Marks		Guidance
6.i	Vertical asymptote is $x = -1$.	**B1**		
	$x^2 + ax - 1 = (x + 1)(x + a - 1) - a$ or $x+1\overline{)x^2+ax-1}$ $x+a-1$	**M1**		By inspection or long division.
	Thus the oblique asymptote is $y = x + a - 1$	**A1**		
		3		
ii	$a^2 + 4 > 0$	**B1**		
		1		
iii	$\frac{(x+1)(2x+a)-(x^2+ax-1)}{(x+1)^2} = 0 \Rightarrow x^2 + 2x + a + 1 = 0$ Discriminant $= 4 - 4(a + 1) < 0$	**M1**		
	Therefore there are no stationary points on C.	**A1**		

		2		
iv		**B1**		Correct y-intercept and asymptotes drawn.
		B1 B1		Each branch correct
		3		

Explanation

6.i

convert to a proper fraction:

$$y = \frac{x^2+ax-1}{x+1} = \frac{x^2+ax+(a-1)-a}{x+1}$$

$$= \frac{(x+1)(x+a-1)-a}{x+1} = x + a - 1 - \frac{a}{x+1}$$ **M1**

Vertical asymptotes:

x+1=0

x=-1 **B1**

Oblique asymptotes:

y=x+a-1 **A1**

ii)

intersection with x-axis: y=0

$$x^2 + ax - 1 = 0$$

$$\Delta = a^2 - 4 \cdot 1 \cdot (-1) = a^2 + 4 > 0$$ **B1**

Thus, the equation will have two distinct roots

So, *C* intersects the *x*-axis twice

iii)

Stationary point: $\frac{dy}{dx} = 0$

$\frac{dy}{dx} = 1 - a(-1)(x+1)^{-2} = 1 + \frac{a}{(x+1)^2} > 0$ always greater than 0 **M1**

There are no stationary points on C **A1**

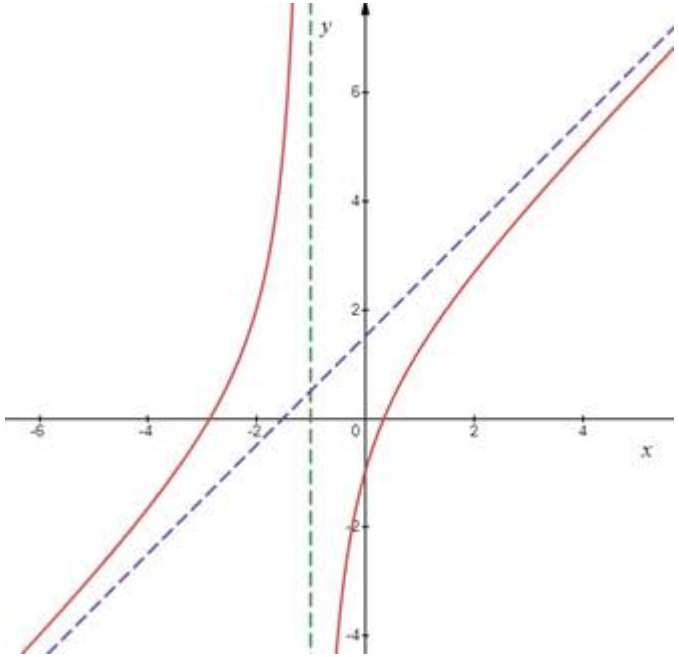
iv)

intersection with y-axis: x=0

y=-1

intersection: (0,-1)

since coefficient of $\frac{a}{x+1}$ term is negative, two branches of curve C are bottom-right and top-left.



B1(Correct y-intercept and asymptotes drawn.)

B1 B1(Each branch correct)

NO. 5

A curve C has equation $y = \frac{ax^2 + x - 1}{x - 1}$, where a is a positive constant.

a. Find the equations of the asymptotes of C. [3]

b. Show that there is no point on C for which $1 < y < 1 + 4a$. [4]

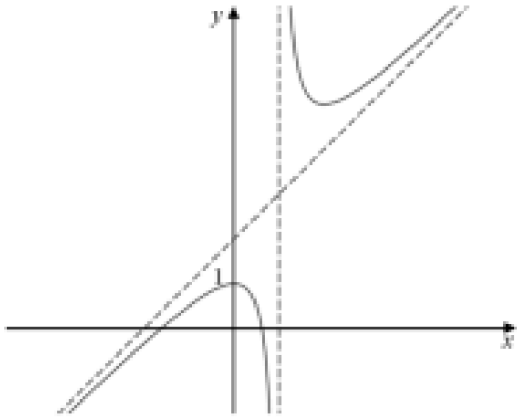
c. Sketch C. You do not need to find the coordinates of the intersections with the axes. [3]

A level FURTHER MATHEMATICS 2022/Summer/Paper1_3/Q3

Mark scheme

Question	Answer	Marks	Guidance
a	$x = 1$	B1	States vertical asymptote.

	$y = \frac{(x-1)(ax+a+1)+a}{x-1} = ax+a+1+\frac{a}{x-1}$	M1	Finds oblique asymptote.
	$y = ax + a + 1$	A1	
		3	
b	$ax^2 + x - 1 = yx - y$ leading to $ax^2 - (y-1)x + (y-1) = 0$	M1 A1	Forms three term quadratic in x.
	$(y-1)^2 - 4a(y-1) < 0$	M1	Correct inequality using discriminant. Must be clear looking for where values of y exist.
	$1 < y < 1 + 4a$	A1	AG Clear method to reach given answer eg $(y-1)(y-1-4a) < 0$
	Alternative method for question 3(b)		
	$ax^2 + x - 1 = yx - y$ leading to $ax^2 - (y-1)x + (y-1) = 0$	M1 A1	Forms three term quadratic in x.
	$(y-1)^2 - 4a(y-1) \geq 0$	M1	Correct inequality using discriminant. Must be clear looking for where values of y exist.
	$1 < y < 1 + 4a$	A1	AG Clear method to reach given answer eg $(y-1)(y-1-4a) \geqslant 0$
	Second alternative method for question 3(b)		
	$\frac{dy}{dx} = \frac{(x-1)(2ax+1)-(ax^2+x-1)}{(x-1)^2} = 0$	M1	
	$x = 0$ or $x = 2$	A1	
	Proves (0, 1) is local maximum and $(2, 4a+1)$ is local minimum.	M1	Either $\frac{d^2y}{dx^2}$ or gradient either side
	Reference to position of asymptote or two distinct branches to justify $1 < y < 1 + 4a$	A1	AG

		4	
c		B1 FT	Asymptotes identified. FT from part **3a**.
		B1	Two branches in correct position relative to correct asymptotes and of correct shape. Good asymptotic behaviour.
		B1	Lower branch in all four quadrants from correct asymptotes.
		3	

NO. 6

The curve C has equation $y = \frac{x^2}{x-3}$.

a. Find the equations of the asymptotes of C. [3]

b. Show that there is no point on C for which 0 < y < 12. [4]

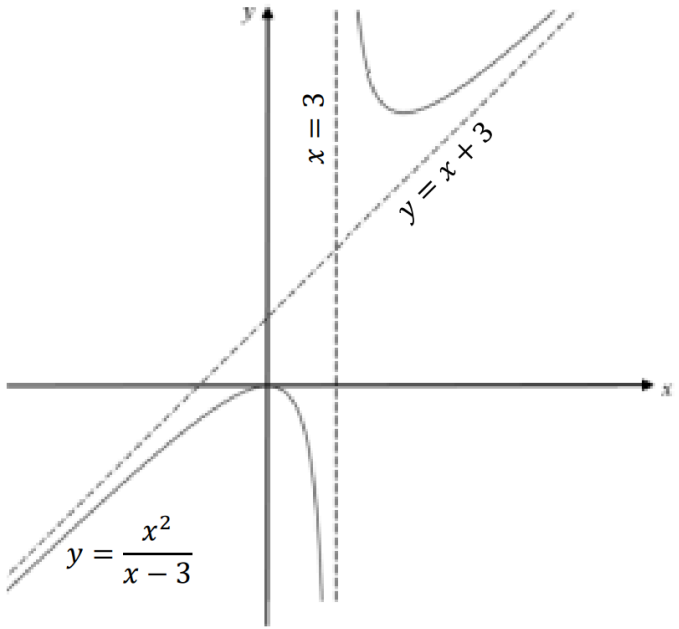
c. Sketch C. [2]

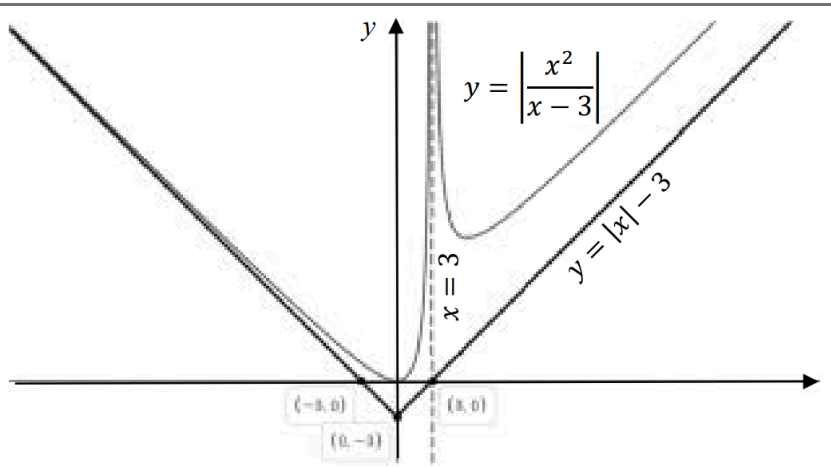
d. i. Sketch the graphs of $y = \left|\frac{x^2}{x-3}\right|$ and $y = |x| - 3$ on a single diagram, stating the coordinates of the intersections with the axes. [4]

ii. Use your sketch to find the set of values of c for which $\left|\frac{x^2}{x-3}\right| \leqslant |x| + c$ has no solution. [1]

A level FURTHER MATHEMATICS 2021/Winter/Paper1_2/Q6

Mark scheme

Question	Answer	Marks	Guidance
a	$x = 3$	B1	States vertical asymptote.
	$y = x + 3 + \frac{9}{x-3}$ leading to $y = x + 3$	M1 A1	Finds oblique asymptote.
		3	
b	$yx - 3y = x^2$ leading to $x^2 - yx + 3y = 0$	M1 A1	Forms quadratic in x.
	$y^2 - 4(3y) < 0$ leading to $y^2 - 12y < 0$	M1	Uses that discriminant is negative.
	$0 < y < 12$	A1	AG
		4	
c		B1	Axes and asymptotes.
		B1	Branches correct.
		2	

d.i		B1 FT	FT from sketch in **(c)**.
		B1	Correct shape at infinity.
		B1	Correct shape of y = $\lvert x \rvert$ − 3.
		B1	Correct intercepts with axes (may be seen on graph).
		4	
d.ii	$c \leqslant -3$	B1	
		1	

NO. 7

The curve C has equation $y = \dfrac{4x+5}{4-4x^2}$.

a. Find the equations of the asymptotes of C. [2]

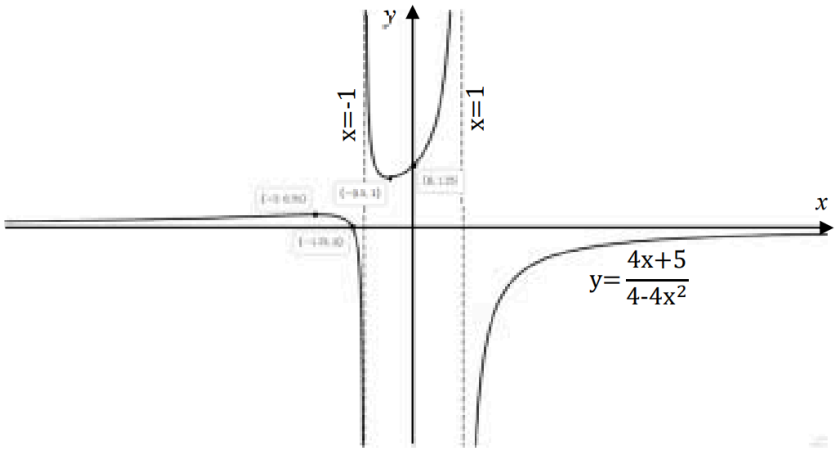
b. Find the coordinates of any stationary points on C. [4]

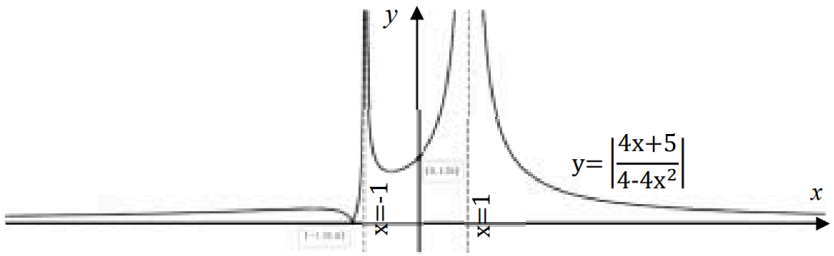
c. Sketch C, stating the coordinates of the intersections with the axes. [3]

d. Sketch the curve with equation $y = \left\lvert \dfrac{4x+5}{4-4x^2} \right\rvert$ and find in exact form the set of values of x for which $4\lvert 4x+5 \rvert > 5\lvert 4-4x^2 \rvert$. [6]

A level FURTHER MATHEMATICS 2021/Winter/Paper1_1/Q7

Mark scheme

Question	Answer	Marks	Guidance
a	$x = 1$, $x = -1$	B1	Vertical asymptotes.
	$y = 0$	B1	Horizontal asymptote.
		2	
b	$\frac{dy}{dx} = \frac{(4-4x^2)(4)-(4x+5)(-8x)}{(4-4x^2)^2}$	M1	Finds $\frac{dy}{dx}$.
	$16x^2 + 40x + 16 = 0$	M1	Sets equal to 0 and forms equation.
	$(-2, \frac{1}{4})$, $(-\frac{1}{2}, 1)$	A1 A1	
		4	
c		B1	Axes and asymptotes.
		B1	Correct shape and position.
	$(-\frac{5}{4}, 0)$, $(0, \frac{5}{4})$	B1	States coordinates of intersections with axes.
		3	

d		B1 FT	FT from sketch in part **(c)**.
		B1	Correct shape at infinity.
	$\frac{4x+5}{4-4x^2}=\frac{5}{4}$ or $\frac{4x+5}{4-4x^2}=-\frac{5}{4}$ $5x^2 + 4x = 0$ or $5x^2 - 4x - 10 = 0$	M2	Finds critical points, award M1 for each case.
	$x=-\frac{4}{5}, \quad x=0 \quad \text{or} \quad x=\frac{2}{5}-\frac{3}{5}\sqrt{6}, \quad x=\frac{2}{5}+\frac{3}{5}\sqrt{6}$	A1	A0 if –1.07, 1.87
	$\frac{2}{5}-\frac{3}{5}\sqrt{6}<x<-\frac{4}{5}, \quad 0<x<\frac{2}{5}+\frac{3}{5}\sqrt{6}$, $x\neq\pm1$	A1 FT	Condone exclusion of x = ±1 from the range.
		6	